# Small endpoints in the single-swap block case

Human Compression project: further bounded L044 proof

We retain the hypotheses and notation of the accepted note *Exact block and orbit deficits at a single crossing*. In particular $G \leq A_n$ is a transitive string C-group, every maximal omission is intransitive, and the unique eligible edge of colour $i$ joins the two $G_i$-orbits. Each orbit now has two blocks, of sizes $k_1, k_2 \geq 4$, and $\alpha = 1$. Put $s = \rho_i$. No assertion below uses a projected intersection property before the projection is proved faithful.

**Lemma 1** (A four-row endpoint forces redundancy)**.** *In the stated setting,*

$$r \leq \left\lfloor \frac{n-1}{2} \right\rfloor .$$

*Thus the accepted single-swap conclusion extends from $k_1, k_2 \geq 5$ to $k_1, k_2 \geq 4$.*

*Proof.* Suppose the bound fails. The exact deficit identity gives $\beta = \gamma = 0$, $r = k_1 + k_2$, and a single simultaneous fixed-point-free block swap $\tau = \rho_{i-1}$, after reversing the order if necessary. The proof of Lemma 14 of the accepted note, up to its endpoint analysis, applies for $k_1, k_2 \geq 4$: the faithful normalized group has the form

$$H = S_a \times S_{b-1}, \qquad \{a, b\} = \{k_1, k_2\}.$$

Here its character is trivial, its right string interval is the diagonal natural $S_a$ moving its row anchor, and its left interval is the diagonal $S_{b-1}$ fixing its row anchor. Both strings have maximal independent length. All symmetric factors used to separate these intervals are centerless, since $a, b \geq 4$; this is the only lower-degree restriction in that part of the argument. The generator $x = \rho_{i-2}$ exists, fixes the two crossing endpoints, preserves all four blocks, centralizes the right interval and the left endpoint omission, and must connect the fixed row of the left interval to its other rows. The accepted argument excludes $b \geq 5$ without requiring $a \geq 5$: the right natural $S_a$ has trivial centralizer for every $a \geq 3$.

It remains to take $b = 4$. Write the rows of its two blocks as $0, 1, 2, 3$ and $0', 1', 2', 3'$, with 0 the crossing endpoint. Its left interval has two generators, necessarily two distinct transpositions on the three nonanchor rows. Relabel them as

$$p = (1\ 2)(1'\ 2'), \qquad q = (2\ 3)(2'\ 3').$$

They occur in the order $p, q, x, \tau, s$. As $x$ centralizes the complete right-hand $S_a$, it is the identity on that original orbit. On the present orbit it centralizes $p$. Its first-block action fixes 0, so is $(1\ 2)^{\epsilon}$. Its second-block action is $(1'\ 2')^{\delta}(0'\ 3')^{\eta}$. If $\eta = 0$, the pair $\{0, 0'\}$ is invariant under $p, q, x, \tau$ and every remaining generator of $G_i$, contradicting this orbit's transitivity. Thus $\eta = 1$. Evenness of $x$ gives $\epsilon + \delta = 1$ modulo two. There are exactly two possibilities:

$$x = (1\ 2)(0'\ 3') \quad \text{or} \quad x = (1'\ 2')(0'\ 3'). \tag{1}$$

The subgroup $\langle p, q, x \rangle$ has, on these eight points, orbits $\{0\}$, $\{1, 2, 3\}$ and $\{0', 1', 2', 3'\}$. Its projections on the latter two are the natural $S_3$ and $S_4$: the second contains $(1'\ 2'), (2'\ 3')$ and, after multiplication by $(1'\ 2')$ if necessary, $(0'\ 3')$. These two natural actions have trivial centralizer and different orbit sizes. Since $s$ commutes with $p, q, x$, it fixes those seven points individually.

Write the rows of the other orbit as $A_0, \dots, A_{a-1}$ and $A'_0, \dots, A'_{a-1}$, with $A_0$ the other crossing endpoint. Only the first generator of the maximal right-hand $S_a$ string moves its row anchor. The anchored classification in Lemma 6 of the accepted note therefore shows that its tail is the natural diagonal $S_{a-1}$ on rows $1, \dots, a-1$. This includes $a = 4$: the first generator may have its extra disjoint transposition, but the last two still generate the full point stabilizer $S_3$. The outer degree-six action is excluded by the fixed anchor, exactly as in the accepted lemma.

The generator $s$ centralizes this tail. Its only possible action on the two $(a-1)$-point orbits is to interchange them row by row, or to fix both pointwise. On the remaining three points $A_0, A'_0, 0$, the unique crossing gives $(A_0\ 0)$, so $A'_0$ is fixed. Fixing the other rows would make $s$ one odd transposition. Hence

$$s = (A_0\ 0) \prod_{j=1}^{a-1} (A_j\ A'_j), \qquad \tau = \prod_{j=0}^{a-1} (A_j\ A'_j)(0\ 0')(1\ 1')(2\ 2')(3\ 3'). \tag{2}$$

In particular $a$ is even, consistently with the sign of the fixed-point-free $\tau$.

With right actions and products applied from left to right, both alternatives in (1) satisfy the exact identity

$$p = s(\tau s x)^5 \tau x. \tag{3}$$

For completeness this identity can be checked without a group-order calculation. On the ten points $A_0, A'_0, 0, 1, 2, 3, 0', 1', 2', 3'$, the permutation $\tau s x$ has cycles

$$(A_0\ A'_0\ 0\ 3'\ 3\ 0') \quad \text{and} \quad \begin{cases} (1\ 1'\ 2\ 2'), & x = (1\ 2)(0'\ 3'), \\ (1\ 2'\ 2\ 1'), & x = (1'\ 2')(0'\ 3'). \end{cases}$$

Its fifth power reverses the six-cycle and leaves the four-cycle as displayed. Multiplication as in (3) gives $(1\ 2)(1'\ 2')$ in each case. On every remaining pair $\{A_j, A'_j\}$, both $s$ and $\tau$ exchange the two points and $x$ is the identity. The word in (3) is therefore the identity there. This proves (3) on all points, for every permitted $a$.

Equation (3) makes the distinguished generator $p$ redundant in the full marking. This contradicts IP, or just independence, and completes the proof. □

**Lemma 2** (A three-row endpoint)**.** *The conclusion also holds with the initial restriction $k_1, k_2 \geq 4$ relaxed to $k_1, k_2 \geq 3$.*

*Proof.* Only a block size three remains after Lemma 1. The faithful normalization from the preceding proof still gives $H = S_a \times S_{b-1}$ with a trivial character, where $a$ is the row orbit whose anchor is moved by the right-hand generator $c$. Indeed the point stabilizer $S_{a-1}$ normally generates $S_a$ already for $a = 3$, and the two-anchor independent-length argument also applies at this endpoint. If $b \geq 4$, the two factors remain centerless and the accepted endpoint analysis applies as before. The case $b = 4$ with $a = 3$ is excluded immediately because $\tau$ would be the product of seven transpositions. Thus assume $b = 3$. Evenness of $\tau$ forces $a$ odd.

The left interval must be nonempty: otherwise $x$ centralizes all of $H$, fixes the unique fixed row of the natural $S_2$ on each three-point block, and leaves its two-point pair invariant under $G_i$. The right interval contains $c$. The projections of the two commuting interval factors onto $S_a$ are commuting normal subgroups whose product is $S_a$. Their intersection is central, hence trivial. Since $S_a$ is directly indecomposable, one projection is trivial; it must be the left one, since the right interval moves the anchor. The left interval is consequently the full central $C_2$ of $H = S_a \times C_2$, generated by one element $p$. The right interval is a complement and hence the graph of the trivial or sign homomorphism $S_a \to C_2$.

The sign graph is impossible. The generator $x$ centralizes the right interval, so is the identity on the original orbit with row group $S_a$. On the three-row orbit it then centralizes the nonidentity $C_2$ action occurring in an odd right-interval element. This fixes its unique fixed row in each block, leaving the anchor pair invariant under $G_i$. Thus the right interval acts trivially on this orbit.

Label its blocks $0,1,2$ and $0',1',2'$. Then $p=(1\ 2)(1'\ 2')$, and $x$ fixes the whole other original orbit. On the first three-point block it fixes 0, and on the second it must move $0'$ to make the orbit transitive. Evenness and a simultaneous exchange of names $1,2$ give

$$x=(1\ 2)(0'\ 1'). \tag{4}$$

The subgroup $\langle p,x\rangle$ has a two-point orbit $\{1,2\}$ and a three-point orbit $\{0',1',2'\}$, on which its images are $S_2$ and $S_3$. Thus $s$, which commutes with both generators, fixes the latter orbit pointwise and on the former may perform $(1\ 2)$.

First take $a\geq 5$. The anchored maximal right interval is the ordinary diagonal Coxeter string, so put $c=(A_0\ A_1)(A_0'\ A_1')$ for its first generator. Its tail is diagonal $S_{a-1}$ on all other rows. Consequently, for some $\epsilon\in\{0,1\}$,

$$\tau=\prod_{j=0}^{a-1}(A_j\ A_j')(0\ 0')(1\ 1')(2\ 2'),\quad s=(A_0\ 0)(1\ 2)\left(\prod_{j=1}^{a-1}(A_j\ A_j')\right)^{\epsilon}. \tag{5}$$

Here $(1\ 2)$ is forced by parity: $a-1$ is even, so either allowed action on those remaining rows is even, while the crossing transposition is odd. Both choices satisfy

$$p=(cs\tau s)^3\tau. \tag{6}$$

To verify it, on the ten points over $A_0,A_1$ and the three-row orbit, $cs\tau s$ has cycles

$$(A_0\ A_1'\ 0\ A_0'\ A_1\ 0'),\qquad (1\ 2'),\qquad (2\ 1').$$

Cubing and multiplying by $\tau$ gives $p$. On every remaining row pair, $cs\tau s$ is $\tau$, so cubing and multiplying by $\tau$ is the identity. This proves (6) uniformly and contradicts independence of $p$.

For clarity the degree-twelve endpoint $a=b=3$ requires one additional centralizer check. The last generator is $t=(A_1\ A_2)(A_1'\ A_2')$, and its centralizer on these four points is $D_8$. Write $s=(A_0\ 0)h(1\ 2)^{\eta}$. Its evenness and involutivity leave precisely

| $\eta$ | $h$ |
|---|---|
| 1 | $1,\quad (A_1\ A_2)(A_1'\ A_2'),\quad (A_1\ A_1')(A_2\ A_2'),\quad (A_1\ A_2')(A_2\ A_1'),$ |
| 0 | $(A_1\ A_2),\quad (A_1'\ A_2').$ |

(7)

For the first four choices, $h$ commutes with the restriction of $\tau$ and fixes the crossing points. Hence $s\tau s$ and the cycle verification of (6) are unchanged. For either of the last two choices, direct multiplication using (4), (7) and the displayed $\tau$ gives instead

$$p=\tau s(\tau x\tau sx)^3\tau x. \tag{8}$$

Indeed $\tau x\tau sx$ has disjoint cycles $(A_0\ 0\ 2\ 1)$, $(0'\ 1'\ 2')$, and $h$. Its cube reverses the four-cycle, kills the three-cycle and retains $h$, giving (8) by multiplication. Thus this is a literal identity on the twelve explicitly named points. All six possibilities again make $p$ redundant. This completes the endpoint and the proof. □

**Scope and attribution.** The source branch is Section 5.1 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for $A_n$*, arXiv:1605.09173v1. The simultaneous-swap and row-normalization framework is the one proved in the accepted block-deficit note, with its full L047/Whiston/extremal dependencies retained. The explicit uniform redundancy words (3), (6) and (8) are the new arguments here. A bounded exploratory permutation calculation suggested that word; the displayed cycle verification is its complete proof, not a sample-based extrapolation or an exhaustive-census premise. The cases with a block size two, the two-independent-swap case, and the remaining L044 principal obligations are not settled by this note. There is no new original, conjecture or formalization claim.